Ansible Mini Project

1. Create an inventory file that includes at least two remote hosts.

```
Dupsy@Dupsy MINGW64 ~/Downloads/ansible/ansible-setup-vagrantfile (main)
$ vagrant status
Current machine states:

master                    running (virtualbox)
node1                     running (virtualbox)
node2                     running (virtualbox)
```

```
[web]
192.168.56.11


[db]
192.168.56.12
~
~
```

```
vagrant@master:~$ ls
inv
vagrant@master:~$ ansible -m ping all -i inv
192.168.56.11 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python"
    },
    "changed": false,
    "ping": "pong"
}
192.168.56.12 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
vagrant@master:~$
```

2. Write an Ansible playbook to perform the following tasks on all hosts specified in the inventory.

```
vagrant@master:~$ sudo touch playbook.yml
vagrant@master:~$ ls
inv  playbook.yml
vagrant@master:~$ sudo vi playbook.yml
```

3. Create a group named devops.

Create the following users and add them to the devops group:

alice

bob

charlie

```yaml
- name: Group, Users, Directories and Files management
  hosts: all
  tasks:
   - name: Create devops group
     group:
       name: devops
       state: present

   - name: Create users and add to group
     user:
       name: "{{ item }}"
       groups: devops
       state: present
     loop:
       - alice
       - bob
       - charlie
```

4.Create the following directories:

/home/devops/projects

/home/devops/logs

Ensure that all files and directories are owned by the devops group and have appropriate permissions:

Directories should have 0755 permissions.

```yaml
- name: Create devops directories with the appropriate permissions
  file:
    path: "{{ item }}"
    state: directory
    owner: root
    group: devops
    mode: "0755"
  loop:
    - /home/devops/projects
    - /home/devops/logs
```

Create the following files:

/home/devops/projects/project1.txt

/home/devops/logs/log1.txt

Ensure that all files and directories are owned by the devops group and have appropriate permissions:

Files should have 0644 permissions.

```yaml
- name: Create files with appropriate permissions
  file:
    path: "{{ item }}"
    state: touch
    owner: root
    group: devops
    mode: "0644"
  loop:
    - /home/devops/projects/project1.txt
    - /home/devops/logs/log1.txt
```

5. Update all system packages using the appropriate package manager (yum for CentOS/RHEL, apt for Debian/Ubuntu).

```yaml
- name: Update system packages
  yum:
     name: '*'
     state: latest
  when: ansible_os_family == "RedHat"

- name: Update system packages
  apt:
    upgrade: yes
    upgrade_cache: yes
  when: ansible_os_family == "Debian"
```

Service Management:

Ensure the httpd (Apache) service is installed, started, and enabled on boot.

```yaml
- name: Install httpd (CentOS)
  yum:
    name: httpd
    state: present
  when: ansible_os_family == "RedHat"

- name: Install apache2 (Debian)
  apt:
    name: apache2
    state: present
  when: ansible_os_family == "Debian"

- name: Ensure httpd service is installed
  service:
    name: httpd
    state: started
    enabled: yes
  when: ansible_os_family == "RedHat"

- name: Ensure apache2 service is installed
  service:
    name: apache2
    state: started
    enabled: yes
  when: ansible_os_family == "Debian"
```

```
vagrant@master:~$ ansible-playbook firstplay.yml -i inv

PLAY [Manage hosts] ************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [192.168.56.11]
ok: [192.168.56.12]

TASK [Create devops group] *****************************************************************************************
ok: [192.168.56.12]
ok: [192.168.56.11]

TASK [Create users and add to group] *******************************************************************************
ok: [192.168.56.12] => (item=alice)
ok: [192.168.56.11] => (item=alice)
ok: [192.168.56.12] => (item=bob)
ok: [192.168.56.11] => (item=bob)
ok: [192.168.56.12] => (item=charlie)
ok: [192.168.56.11] => (item=charlie)

TASK [Create devops directories with the appropriate permissions] ***************************************************
ok: [192.168.56.12] => (item=/home/devops/projects)
ok: [192.168.56.11] => (item=/home/devops/projects)
- name: Manage hosts
vagrant@master:~$ ansible-playbook firstplay.yml -i inv

PLAY [Manage hosts] ************************************************************************************************
```

```
TASK [Gathering Facts] *********************************************************************************************
ok: [192.168.56.11]
ok: [192.168.56.12]

TASK [Create devops group] *****************************************************************************************
ok: [192.168.56.12]
ok: [192.168.56.11]

TASK [Create users and add to group] *******************************************************************************
ok: [192.168.56.12] => (item=alice)
ok: [192.168.56.11] => (item=alice)
ok: [192.168.56.12] => (item=bob)
ok: [192.168.56.11] => (item=bob)
ok: [192.168.56.12] => (item=charlie)
ok: [192.168.56.11] => (item=charlie)

TASK [Create devops directories with the appropriate permissions] ***************************************************
ok: [192.168.56.12] => (item=/home/devops/projects)
ok: [192.168.56.11] => (item=/home/devops/projects)
ok: [192.168.56.12] => (item=/home/devops/logs)
ok: [192.168.56.11] => (item=/home/devops/logs)

TASK [Create files with appropriate permissions] *******************************************************************
changed: [192.168.56.12] => (item=/home/devops/projects/project1.txt)
changed: [192.168.56.11] => (item=/home/devops/projects/project1.txt)
changed: [192.168.56.12] => (item=/home/devops/logs/log1.txt)
changed: [192.168.56.11] => (item=/home/devops/logs/log1.txt)
```

```
TASK [Create files with appropriate permissions] ***********************************************************************
changed: [192.168.56.12] => (item=/home/devops/projects/project1.txt)
changed: [192.168.56.11] => (item=/home/devops/projects/project1.txt)
changed: [192.168.56.12] => (item=/home/devops/logs/log1.txt)
changed: [192.168.56.11] => (item=/home/devops/logs/log1.txt)

TASK [Update system packages] ******************************************************************************************
skipping: [192.168.56.12]
ok: [192.168.56.11]

TASK [Update system packages] ******************************************************************************************
skipping: [192.168.56.11]
changed: [192.168.56.12]

TASK [Install httpd (CentOS)] ******************************************************************************************
skipping: [192.168.56.12]
changed: [192.168.56.11]

TASK [Install apache2 (Debian)] ****************************************************************************************
skipping: [192.168.56.11]
changed: [192.168.56.12]

TASK [Ensure httpd service is installed] *******************************************************************************
skipping: [192.168.56.12]
changed: [192.168.56.11]
```

```
ok: [192.168.56.11]

TASK [Update system packages] ******************************************************************************************
skipping: [192.168.56.11]
changed: [192.168.56.12]

TASK [Install httpd (CentOS)] ******************************************************************************************
skipping: [192.168.56.12]
changed: [192.168.56.11]

TASK [Install apache2 (Debian)] ****************************************************************************************
skipping: [192.168.56.11]
changed: [192.168.56.12]

TASK [Ensure httpd service is installed] *******************************************************************************
skipping: [192.168.56.12]
changed: [192.168.56.11]

TASK [Ensure apache2 service is installed] *****************************************************************************
skipping: [192.168.56.11]
ok: [192.168.56.12]

PLAY RECAP *************************************************************************************************************
192.168.56.11              : ok=8    changed=3    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
192.168.56.12              : ok=8    changed=3    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0

vagrant@master:~$
```

Create a script named copy_passwd.sh on the control node with the following

content:

#!/bin/bash

cat /etc/passwd > passwd_copy.txt

echo "Contents of /etc/passwd copied to passwd_copy.txt"

```
vagrant@master:~$ touch copy_passwd.sh
vagrant@master:~$ sudo vi copy_passwd.sh
vagrant@master:~$
```

```
#!/bin/bash

cat /etc/passwd > passwd_copy.txt
echo "Contents of /etc/passwd copied to passwd_copy.txt"
~
~
~
~
~
~
```

Ensure the script is executable by running the following command on the control node:

chmod +x copy_passwd.sh

```
vagrant@master:~$ touch copy_passwd.sh
vagrant@master:~$ sudo vi copy_passwd.sh
vagrant@master:~$ sudo vi copy_passwd.sh
vagrant@master:~$
vagrant@master:~$
vagrant@master:~$
vagrant@master:~$ ll
total 68
drwxr-xr-x 6 vagrant vagrant  4096 Jun  8 06:09 ./
drwxr-xr-x 4 root    root     4096 Jun  1 12:50 ../
drwx------ 4 vagrant vagrant  4096 Jun  1 13:15 .ansible/
-rw------- 1 vagrant vagrant  7903 Jun  8 05:06 .bash_history
-rw-r--r-- 1 vagrant vagrant   220 Jun  7  2023 .bash_logout
-rw-r--r-- 1 vagrant vagrant  3771 Jun  7  2023 .bashrc
drwx------ 2 vagrant vagrant  4096 Jun  1 12:50 .cache/
drwx------ 3 vagrant vagrant  4096 Jun  1 12:50 .gnupg/
-rw-r--r-- 1 vagrant vagrant   807 Jun  7  2023 .profile
drwx------ 2 vagrant vagrant  4096 Jun  1 16:55 .ssh/
-rw------- 1 root    root    10165 Jun  8 06:09 .viminfo
-rw-rw-r-- 1 vagrant vagrant   104 Jun  8 06:09 copy_passwd.sh
-rw-r--r-- 1 root    root     1450 Jun  8 05:52 firstplay.yml
-rw-r--r-- 1 root    root       37 Jun  6 12:54 inv
vagrant@master:~$
```

```
vagrant@master:~$
vagrant@master:~$
vagrant@master:~$ chmod +x copy_passwd.sh
vagrant@master:~$ ll
total 68
drwxr-xr-x 6 vagrant vagrant  4096 Jun  8 06:09 ./
drwxr-xr-x 4 root    root     4096 Jun  1 12:50 ../
drwx------ 4 vagrant vagrant  4096 Jun  1 13:15 .ansible/
-rw------- 1 vagrant vagrant  7903 Jun  8 05:06 .bash_history
-rw-r--r-- 1 vagrant vagrant   220 Jun  7  2023 .bash_logout
-rw-r--r-- 1 vagrant vagrant  3771 Jun  7  2023 .bashrc
drwx------ 2 vagrant vagrant  4096 Jun  1 12:50 .cache/
drwx------ 3 vagrant vagrant  4096 Jun  1 12:50 .gnupg/
-rw-r--r-- 1 vagrant vagrant   807 Jun  7  2023 .profile
drwx------ 2 vagrant vagrant  4096 Jun  1 16:55 .ssh/
-rw------- 1 root    root    10165 Jun  8 06:09 .viminfo
-rwxrwxr-x 1 vagrant vagrant   104 Jun  8 06:09 copy_passwd.sh*
-rw-r--r-- 1 root    root     1450 Jun  8 05:52 firstplay.yml
-rw-r--r-- 1 root    root       37 Jun  6 12:54 inv
vagrant@master:~$
```

Use an Ansible ad-hoc command to copy and execute the script on all target

servers.

```
#!/bin/bash
vagrant@master:~$ ansible all -i inv -m copy -a 'src=copy_passwd.sh dest=/home/vagrant/copy_passwd.sh mode=775'
192.168.56.12 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": true,
    "checksum": "8c69cc4494c653597bd2cca30d68542a6b67bb2d",
    "dest": "/home/vagrant/copy_passwd.sh",
    "gid": 1000,
    "group": "vagrant",
    "md5sum": "bdd6830818691bda5cd2a5e7771657cd",
    "mode": "0775",
    "owner": "vagrant",
    "size": 104,
    "src": "/home/vagrant/.ansible/tmp/ansible-tmp-1717827983.15-2433-171435500283968/source",
    "state": "file",
    "uid": 1000
}
192.168.56.11 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python"
    },
    "changed": true,
    "checksum": "8c69cc4494c653597bd2cca30d68542a6b67bb2d",
    "dest": "/home/vagrant/copy_passwd.sh",
    "gid": 1000,
    "group": "vagrant",
    "md5sum": "bdd6830818691bda5cd2a5e7771657cd",
    "mode": "0775",
    "owner": "vagrant",
    "secontext": "unconfined_u:object_r:user_home_t:s0",
    "size": 104,
    "src": "/home/vagrant/.ansible/tmp/ansible-tmp-1717827982.21-2431-214622811895613/source",
    "state": "file",
    "uid": 1000
}
vagrant@master:~$
```

```
vagrant@master:~$ ansible all -i inv -m shell -a '/home/vagrant/copy_passwd.sh'
192.168.56.11 | CHANGED | rc=0 >>
Contents of /etc/passwd copied to passwd_copy.txt
192.168.56.12 | CHANGED | rc=0 >>
Contents of /etc/passwd copied to passwd_copy.txt
vagrant@master:~$
```